

Milestone 2: Multi - Agent Workflow

Agent Description:

The Meal Planning Agent is an agentic AI solution that can assist users in generating personalized weekly meal plans for users. The system prompts users to input all of this information in the interface, and then generates a cohesive, 7 day plan that lists specific meals for each day, along with nutritional information such as protein levels, calories, and anything else the user requests. In addition, the system can also provide the user with some budget friendly grocery shopping tips, as well as general guidance on health and diet

Agent Roles + Input/Output:

- **Data Aggregator Agent**
 - Role: Collects all user information for meal planning
 - Output: Organized user preferences
 - Input: Dietary preferences, schedule, inventory of ingredients
- **Meal Planner Agent**
 - Role: Creates a meal plan based on the user's needs
 - Output: A structured meal plan
 - Input: User information from the data aggregator
- **Critic Agent**
 - Role: Reviews the meal plan to make sure it fits the user's needs
 - Output: Feedback to improve meal plan
 - Input: Generated Meal plan
- **Presenter Agent**
 - Role: Formats and presents meal plan to user in a functional way
 - Output: Final meal plan to show to user
 - Input: Finalized meal plan with all user requirements

Agent Coordination/Communication Mechanisms:

- This is a fixed pipeline
 - Data Aggregator Agent → Meal Planner Agent → Critic Agent → Presenter Agent
 - The communication of this pipeline is directly synchronous as it requires one agent to complete its tasks before the next agent can complete its task. Essentially, this means each agent has a specialized role in the functionality. The data aggregator agent starts off the pipeline and requests input from the user to collect all the requirements needed for further functions. This information is then passed to the meal planner agent, which then begins to develop a comprehensive meal plan based on the requirements/inputs. Once this is

complete, the critic agent reviews the draft from the meal planner agent, to ensure that what is being done is strictly what the user asked for and provided the agent with. Finally, once this task is complete, the critic agent communicates to the presenter agent that the requirements are verified and the presenter agent shows the user a function UI with a comprehensive summary of the plan, including dietary information and restrictions.

- I will incorporate external APIs in the upcoming milestone.

Failure + Quality Handling:

- One way the system handles failure is with the critic agent. This ensures that only the items that the user listed are found in the meal plans, nothing more
- The data aggregator agent allows the user to put in very specific information, including intolerances, allergies, and any necessary provisions to protect the user.
- The presenter agent prevents any unnecessary UI designs, and ensures that the program does not fail because of a dysfunctional “aesthetic”.

Code:

Because I used Opal, a low-code, no-code platform, I do not have any code.

Risks	Mitigation Plan	
	Design & development of agent	Agent integration into processes
Adversarial attacks	-input adversarial training -anomaly detection for any unusual dietary requests -adversarial training	-verification before the presenter agent displays anything -decision auditing across the pipeline
Bias and discrimination	-use diverse training data so meal recommendations don't prefer any specific group -create accountability frameworks for autonomous actions	-bias detection so the critic agent can stop it before it gets to the presenter -decision auditing for outputs
Misinformation and hallucinations	-provide real time verification for information with external sources(<u>api's</u> for reliable websites) -implement uncertainty-aware <u>decision making</u> processes	-multi step verification for very specific restrictions(allergies) -human input to prevent serious situations related to health issues
Data privacy violations	-robust data governance to prevent PI from being collected in the aggregator agent -automatic privacy aware decision protocols	-real time privacy monitoring guides to ensure no unauthorized sharing of user data
Intellectual property risks	-data auditing	-ensure legal compliance to prevent data leakage or similar issues -confirm copyright

